

Solutions for Homework Assignment #1

Answer to Question 1.

a. In what follows **placement** is an arrangement of all the given books on shelves so that (a) the books appear in the given order on successive shelves, and (b) the sum of the depths of the books on each shelf does not exceed L . A placement **optimal** if it minimizes the sum of the empty spaces on the shelves it uses.

Let P be the placement produced by the greedy algorithm, and m be the number of shelves used in that placement. We will show:

Claim. For each s , $0 \leq s \leq m$, there is an optimal placement that is identical to P through shelf s .

The optimality of the algorithm follows from this claim by taking $s = m$.

We prove the claim by induction on s . The basis, $s = 0$, is obvious. For the induction step, let s be an arbitrary integer such that $0 \leq s < m$, and suppose that there is an optimal placement P_s that is identical to P through shelf s . We will show that there is an optimal placement P_{s+1} that is identical to P through shelf $s + 1$.

P_{s+1} obtained from P_s as follows: Because P_s is identical to P through shelf s , shelf $s + 1$ of P_s contains a prefix of the sequence of books on shelf $s + 1$ of P (otherwise the algorithm would have put more books on shelf $s + 1$ of P). If this prefix actually consists of all the books on shelf $s + 1$ of P , we take $P_{s+1} = P_s$ and we are done: P_{s+1} is an optimal placement that agrees with P through shelf $s + 1$. Otherwise, there is a nonempty suffix of the sequence of books on shelf $s + 1$ of P that are not on shelf $s + 1$ of P_s . These must all be on shelf $s + 2$ of P_s (otherwise, P_s would not be optimal: a placement with less empty space could be produced by moving all the books of shelf $s + 2$ to shelf $s + 1$ and not using shelf $s + 2$ at all). We obtain P_{s+1} from P_s by transferring these books to shelf $s + 1$. Thus, P_{s+1} is identical to P through shelf $s + 1$. It remains to argue that P_{s+1} is optimal. This is because P_s is optimal, and the sum of empty spaces on the shelves of P_{s+1} is equal to that of P_s : the additional empty space created on shelf $s + 2$ by the transfer of books is offset exactly by the reduced empty space on shelf $s + 1$.

Note: This is like a “promising set” argument. The solution is a sequence (of books on each shelf), rather than a set, so we have a “promising prefix” rather than a “promising set”.

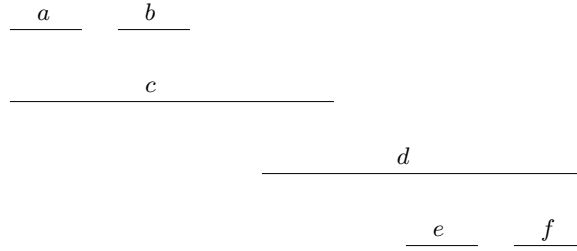
b. The algorithm does not minimize the maximum empty space on any used shelf, as the following counterexample shows: Suppose $d_1 = 2$, $d_2 = 1$, $d_3 = 1$ and $L = 3$.

The algorithm places books 1 and 2 on shelf 1 leaving empty space $3 - (2 + 1) = 0$, and book 3 on shelf 2 leaving empty space $3 - 1 = 2$. So, the maximum empty space on any shelf is 2.

If we place book 1 on shelf 1, and books 2 and 3 on shelf 2, the empty space on both shelves is $3 - 2 = 1$, which is smaller than the maximum empty space in the placement produced by the algorithm.

Answer to Question 2.

a. This is false as shown by the following example.



The unique minimum cardinality cover in this case is $\{c, d\}$, but these two intervals overlap.

b. **ALGORITHM.** The idea is to consider the intervals in non-decreasing finish time and keep track of the cover $\mathcal{C}$ “so far” (initially empty). When interval I is considered, if it is not already covered by $\mathcal{C}$, we add to $\mathcal{C}$ an interval that covers I and has maximum finish time. Here is the algorithm in pseudocode:

```

COVER( $\mathcal{I}$ )
1   $\mathcal{L} :=$  list of intervals in  $\mathcal{I}$  sorted by non-decreasing finish time
2   $\mathcal{C} := \emptyset$ 
3  while  $\mathcal{L} \neq \emptyset$  do
4       $I :=$  first interval in  $\mathcal{L}$ 
5      if  $I$  does not intersect any interval in  $\mathcal{C}$  then
6           $I' :=$  interval in  $\mathcal{L}$  that intersects  $I$  and has maximum finish time (ties broken arbitrarily)
7           $\mathcal{C} := \mathcal{C} \cup \{I'\}$ 
8          remove  $I$  from  $\mathcal{L}$ 
9  return  $\mathcal{C}$ 

```

The intuition why this is correct is that we “cover” each interval I of $\mathcal{I}$ using the interval that intersects I and extends as far to the right as possible in the hope that this will also cover the maximum number of other intervals, thereby minimizing the number of intervals needed to cover all intervals. This intuition is more or less correct, but it is incomplete: for example, it does not explain why it is important to sort the intervals by non-decreasing *finish* time as opposed to, say, *start* time. To make sure you understand the proof thoroughly you should: (a) show that the algorithm would not be correct if we sorted the intervals by non-decreasing start time; and (b) pinpoint the place(s) in the proof below where the fact that intervals are sorted by finish time is relevant. When we add a new interval I' to $\mathcal{C}$ it may seem tempting to delete from $\mathcal{L}$ all intervals covered by I' , but this would be wrong! Do you see why?

CORRECTNESS. First we note that the algorithm terminates, since in each iteration one interval is removed from $\mathcal{I}$ (line 8), so after n iterations $\mathcal{I}$ is empty.

To prove that the set of intervals $\mathcal{C}$ returned by the algorithm is optimal, i.e., it is a minimum cardinality cover of $\mathcal{I}$, we use the “promising set” approach. Specifically we prove:

Promising Set Lemma. *At the end of each iteration, (a) $\mathcal{C}$ is a cover of the set of intervals removed from $\mathcal{L}$ (i.e., the set of intervals in $\mathcal{I} - \mathcal{L}$), and (b) $\mathcal{C}$ is contained in an optimal cover of $\mathcal{I}$.*

Proof. The proof is by induction on the iteration number. Let $\mathcal{C}_t$ and $\mathcal{L}_t$ be the set of intervals in variables $\mathcal{C}$ and $\mathcal{L}$ at the end of iteration t . (As usual “the end of iteration 0” refers to the point just before the first iteration of the loop.)

The basis is trivially true, since $\mathcal{C}_0 = \emptyset$ and nothing has been removed from $\mathcal{L}$. Suppose the invariant holds after iteration t , and let $\mathcal{C}_t^*$ be an optimal set that contains $\mathcal{C}_t$. We will now prove that, if iteration $t+1$ exists, the invariant remains true after iteration $t+1$; that is, (a) $\mathcal{C}_{t+1}$ is a cover of the set of intervals in $\mathcal{I} - \mathcal{L}_{t+1}$, and (b) $\mathcal{C}_{t+1}$ is contained in some optimal cover $\mathcal{C}_{t+1}^*$ of $\mathcal{I}$.

Part (a) is true by the induction hypothesis for (a) and the fact that the interval I deleted from $\mathcal{L}$ in iteration $t + 1$ either intersects an interval that is already in $\mathcal{C}_t$ (and therefore in $\mathcal{C}_{t+1}$), or intersects the interval I' added to $\mathcal{C}$ in iteration $t + 1$.

For part (b), if no interval is added to $\mathcal{C}$ in iteration $t + 1$ (i.e., the condition in line 5 is false) then we simply take $\mathcal{C}_{t+1}^* = \mathcal{C}_t^*$ and we are done by the induction hypothesis. So, suppose that interval $I' = [s', f']$ is added to $\mathcal{C}$ (line 7) and interval $I = [s, f]$ is the interval deleted from $\mathcal{L}$ (line 8) in iteration $t + 1$. There are two cases:

CASE 1. $I' \in \mathcal{C}_t^*$. Then we take $\mathcal{C}_{t+1}^* = \mathcal{C}_t^*$ and we are done by the induction hypothesis.

CASE 2. $I' \notin \mathcal{C}_t^*$. Then there is some interval, say $I'' = [s'', f'']$, in $\mathcal{C}_t^*$ that intersects I (otherwise $\mathcal{C}_t^*$ would not be a cover of $\mathcal{I}$, let alone an optimal one). In this case, we take $\mathcal{C}_{t+1}^* = (\mathcal{C}_t^* - \{I''\}) \cup \{I'\}$; that is, we replace I'' by I' in $\mathcal{C}_t^*$. We claim that

$$\mathcal{C}_{t+1}^* \text{ is an optimal cover of } \mathcal{I} \text{ that contains } \mathcal{C}_{t+1}. \quad (*)$$

To prove this, we first note that $I'' \notin \mathcal{C}_t$, for if it was then I would already be covered and I' would not be added to $\mathcal{C}$ in iteration $t + 1$. By induction hypothesis, $\mathcal{C}_t \subseteq \mathcal{C}_t^*$, and since $I'' \notin \mathcal{C}_t$, $\mathcal{C}_t \subseteq \mathcal{C}_t^* - \{I''\}$. So, $\mathcal{C}_{t+1} = \mathcal{C}_t \cup \{I'\} \subseteq (\mathcal{C}_t^* - \{I''\}) \cup \{I'\} = \mathcal{C}_{t+1}^*$. So, $\mathcal{C}_{t+1}^*$ contains $\mathcal{C}_{t+1}$. Furthermore, $\mathcal{C}_{t+1}^*$ contains the same number of intervals as the optimal cover $\mathcal{C}_t^*$ of $\mathcal{I}$. So, to prove (*) it remains to show that $\mathcal{C}_{t+1}^*$ is a cover of $\mathcal{I}$. Since $\mathcal{C}_{t+1}^*$ differs from $\mathcal{C}_t^*$, which is a cover of $\mathcal{I}$, only in that it replaces I'' by I' , it suffices to prove that

$$\text{If } J \in \mathcal{I} \text{ intersects } I'' \text{ then it also intersects some interval in } \mathcal{C}_{t+1}^*. \quad (**)$$

To prove (**), let $J = [s_J, f_J]$ be any interval in $\mathcal{I}$ that intersects I'' . If it also intersects I' , we are done (since $I' \in \mathcal{C}_{t+1}^*$). So, suppose that J does not intersect I' . Therefore, either (i) $f' < s_J$ or (ii) $f_J < s'$.

- In case (i), we have $f'' \leq f' < s_J$ (the first inequality is because we consider intervals in $\mathcal{I}$ in order of finish time and because I' is chosen as an interval in $\mathcal{L}_t$ that intersects I and has maximum finish time). This contradicts that I'' intersects J , so this case is impossible.
- In case (ii), we have $f_J < s' \leq f$ (the second inequality is because I' intersects I). By part (a) of the induction hypothesis, $\mathcal{C}_t$ covers all intervals that are deleted from $\mathcal{L}_t$, and since $f_J < f$ and we consider intervals by non-decreasing finish time, J is deleted from $\mathcal{L}_t$, so J is covered by $\mathcal{C}_t$. We have shown above that $\mathcal{C}_t \subseteq \mathcal{C}_{t+1}^*$, so J is covered by $\mathcal{C}_{t+1}^*$. This completes the proof of (**), and therefore of the induction step for part (b). $\square$

It is now easy to see that the Promising Set Lemma implies that the set $\mathcal{C}$ returned by the algorithm is an optimal cover of $\mathcal{I}$: By the exit condition of the loop in line 3, all intervals in $\mathcal{I}$ have been deleted from $\mathcal{L}$, so by part (a) of the lemma, $\mathcal{C}$ is a cover of $\mathcal{I}$. By part (b) of the lemma, $\mathcal{C}$ is contained in a minimum cardinality cover of $\mathcal{I}$, so it must itself be a minimum cardinality cover of $\mathcal{I}$.

RUNNING TIME. Sorting of the n intervals by finish time can be done in $O(n \log n)$ time. As argued above, the loop terminates after at most n iterations. Determining whether I intersects any interval in $\mathcal{C}$ (line 5) and finding the interval I' (line 6) can each be done in $O(n)$ time. So each iteration of the loop takes $O(n)$ time, and the entire algorithm takes $O(n^2)$ time.